Detailed Theory: Static Routing, Dynamic Routing, and Routing Protocols

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1. Static Routing

What is Static Routing?

- Static routing is a method where network routes are manually configured and entered into the router by the network administrator.
- Routes do not change unless manually updated.
- Used in small networks or simple topologies.
- Ideal when routes are stable and predictable.

Characteristics:

- **Manual Configuration:** Each route is defined by the admin.
- **No Overhead:** No routing updates or computations.
- **No Automatic Failover:** If a route fails, manual intervention is required.
- **Low CPU and Memory Use:** Because no routing algorithm runs.
- **Secure:** Less susceptible to routing attacks since no route advertisement.

Example Static Route Command (Cisco):

scss

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Router(config)# ip route [destination_network] [subnet_mask] [next_hop_ip or exit_interface]

Example:

nginx

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ip route 192.168.2.0 255.255.255.0 192.168.1.2

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2. Dynamic Routing

What is Dynamic Routing?

- Dynamic routing allows routers to automatically discover network destinations and adjust routes as the network topology changes.

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Uses routing protocols to exchange routing information between routers.

- Essential for large or frequently changing networks.

Characteristics:

- **Automatic Route Discovery and Maintenance:** Routers communicate and share routes.
- **Adaptability:** Automatically recalculates routes when topology changes.
- **Routing Overhead:** Uses bandwidth and CPU for routing updates and computations.
- **Supports Scalability:** Ideal for large and complex networks.
- **Routing Loops Handling:** Uses algorithms to prevent routing loops.

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3. Routing Protocols

Routing protocols define the rules and algorithms used by routers to exchange routing information.

Classification of Routing Protocols:

Type	Description	Examples
Distance Vector	Routers share routing tables periodically	RIP, IGRP
Link-State	Routers share information about neighbors	OSPF, IS-IS
Hybrid	Combines features of both Distance Vector and Link-State	EIGRP

3.1 Distance Vector Routing Protocols

- Send full routing tables to neighbors at regular intervals.

- Use metrics like hop count.
- Simpler but slower convergence and prone to routing loops.
- Examples: RIP (Routing Information Protocol), IGRP (Interior Gateway Routing Protocol).

3.2 Link-State Routing Protocols

- Each router builds a map of the entire network topology.
- Exchanges Link State Advertisements (LSAs) only when topology changes.
- Faster convergence and scalable.
- Examples: OSPF (Open Shortest Path First), IS-IS.

3.3 Hybrid Routing Protocols

- Use distance vector algorithms with some link-state features.
- Support rapid convergence and scalability.
- Example: EIGRP (Enhanced Interior Gateway Routing Protocol).

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4. Common Routing Protocols Overview

Protocol	Type	Metric Used	Max Hop Count	Convergence Time	Use Case
RIP	Distance Vector	Hop count	15	Slow	Small networks
OSPF	Link-State	Cost (bandwidth)	Unlimited	Fast	Large enterprise networks
EIGRP	Hybrid	Bandwidth, delay	255	Fast	Cisco proprietary, scalable
BGP	Path Vector	AS path length	N/A	Slow	Internet backbone routing

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5. Advantages and Disadvantages

Routing Type	Advantages	Disadvantages
Static Routing	Simple, no overhead, secure	No automatic failover, difficult to scale
Dynamic Routing	Automatic route management, scalable	Higher CPU/bandwidth usage, complex

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6. When to Use Static vs Dynamic Routing

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Static Routing: Small networks, stub networks, backup routes, or security sensitive paths.

- **Dynamic Routing:** Large, complex networks where routes change frequently.